

Binärsystem

Stefan Mangold
& Michael Geuze



$$\begin{array}{rcl}
 1) & 18 : 2 = 9 & 0 \text{ R} \\
 & 9 : 2 = 4 & 1 \text{ R} \\
 & 4 : 2 = 2 & 0 \text{ R} \\
 & 2 : 2 = 1 & 0 \text{ R} \\
 & 1 : 2 = 0 & 1 \text{ R}
 \end{array}
 \uparrow$$

$$18_{10} = 0001 \ 0010_2$$

$$\begin{array}{rcl}
 2) & 179 : 2 = 89 & 1 \text{ R} \\
 & 89 : 2 = 44 & 1 \text{ R} \\
 & 44 : 2 = 22 & 0 \text{ R} \\
 & 22 : 2 = 11 & 0 \text{ R} \\
 & 11 : 2 = 5 & 1 \text{ R} \\
 & 5 : 2 = 2 & 1 \text{ R} \\
 & 2 : 2 = 1 & 0 \text{ R} \\
 & 1 : 2 = 0 & 1 \text{ R}
 \end{array}$$

$$179_{10} = 1011 \ 0011_2$$

$$\begin{array}{rcl}
 3) & 54351 : 2 = 27175 & 1 \text{ R} \\
 & 27175 : 2 = 13587 & 1 \text{ R} \\
 & 13587 : 2 = 6793 & 1 \text{ R} \\
 & 6793 : 2 = 3396 & 1 \text{ R} \\
 & 3396 : 2 = 1698 & 0 \text{ R} \\
 & 1698 : 2 = 849 & 0 \text{ R} \\
 & 849 : 2 = 424 & 1 \text{ R} \\
 & 424 : 2 = 212 & 0 \text{ R} \\
 & 212 : 2 = 106 & 0 \text{ R} \\
 & 106 : 2 = 53 & 0 \text{ R} \\
 & 53 : 2 = 26 & 1 \text{ R} \\
 & 26 : 2 = 13 & 0 \text{ R} \\
 & 13 : 2 = 6 & 1 \text{ R} \\
 & 6 : 2 = 3 & 0 \text{ R} \\
 & 3 : 2 = 1 & 1 \text{ R} \\
 & 1 : 2 = 0 & 1 \text{ R}
 \end{array}$$

$$54351_{10} = 1101 \ 0100 \ 0100 \ 1111_2$$

$$0,625_{10} = 0,101_2$$

$$3, 3_{10} = 0011, 011001$$

$$2,21828_{10} = 0010,1011\ 0111\ 1111$$

$$11001_2 = 25_{10}$$

$$11001101_2 = 413_{10}$$

$$11101111101_2 = 1917_{10}$$

$$10) \quad 0,1011$$

$$z = 4$$

$$0,1011_2 = \frac{11}{2^4} = \frac{11}{16} = 0,6875$$

$$\Rightarrow 1011$$

$$8+2+1 = 11_{10}$$

$$0,1011_2 = 0,6875_{10}$$

$$11) \quad 0,11011001_2$$

$$z = 8$$

$$0,11011001_2 = \frac{212}{2^8} = \frac{212}{256} = 0,84765625$$

$$\Rightarrow 11011001$$

$$128 + 64 + 16 + 8 + 1 = 212$$

$$0,11011001_2 = 0,84765625_{10}$$

$$12) \quad 10001,0001111101$$

$$\begin{array}{r} 10001_2 = 17_{10} \\ 16 + 1 \end{array}$$

$$0001111101_2 = \frac{125}{2^{10}} = \frac{125}{1024} = 0,1220703125_{10}$$

$$111101$$

$$64 + 32 + 16 + 8 + 4 + 1 = 125_{10}$$

$$10001,0001111101_2 = 17,1220703125_{10}$$

$$\Rightarrow z = 10$$